

DATA VISUALIZATION SCREENSHOT

TASKS:

Overview

Problem Statement

Taxi services generate large amounts of trip data daily, which can be used to understand patterns and improve operations. However, raw data often contains missing values and is difficult to interpret without proper analysis.

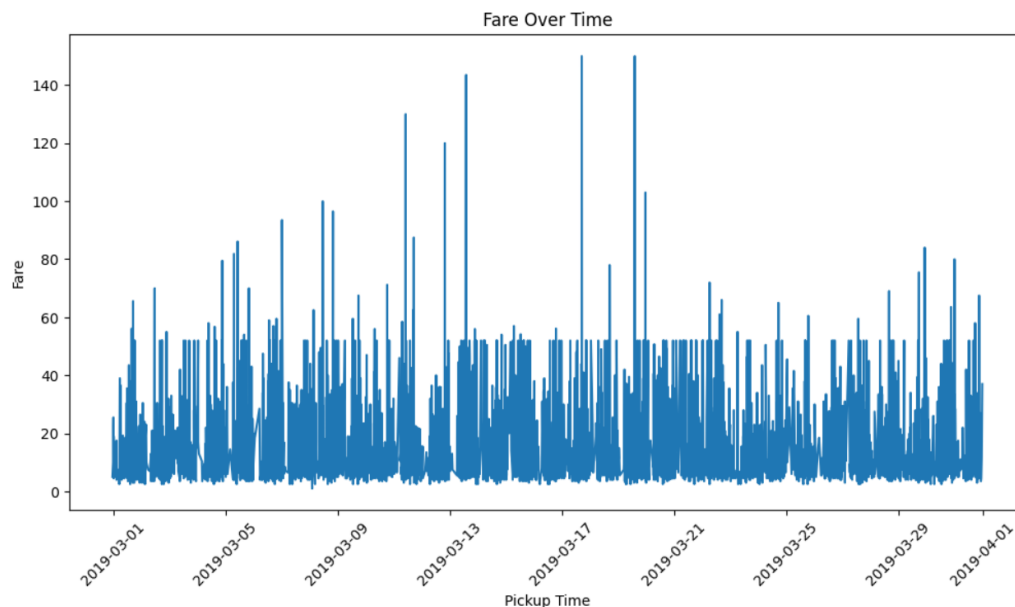
In this project, you will act as a **data analyst** to clean, analyze, and visualize taxi data. You will handle missing values, explore trends, and create visualizations to gain insights into fare, distance, and customer behavior using **Pandas, Matplotlib, and Seaborn**.

1) Visualizations using Matplotlib/Pandas Plot:

★ Line Chart

Plot a line chart to visualize the fare over time, using the pickup timestamp as the x-axis and fare as the y-axis. Ensure the pickup column is converted to a datetime format before plotting.

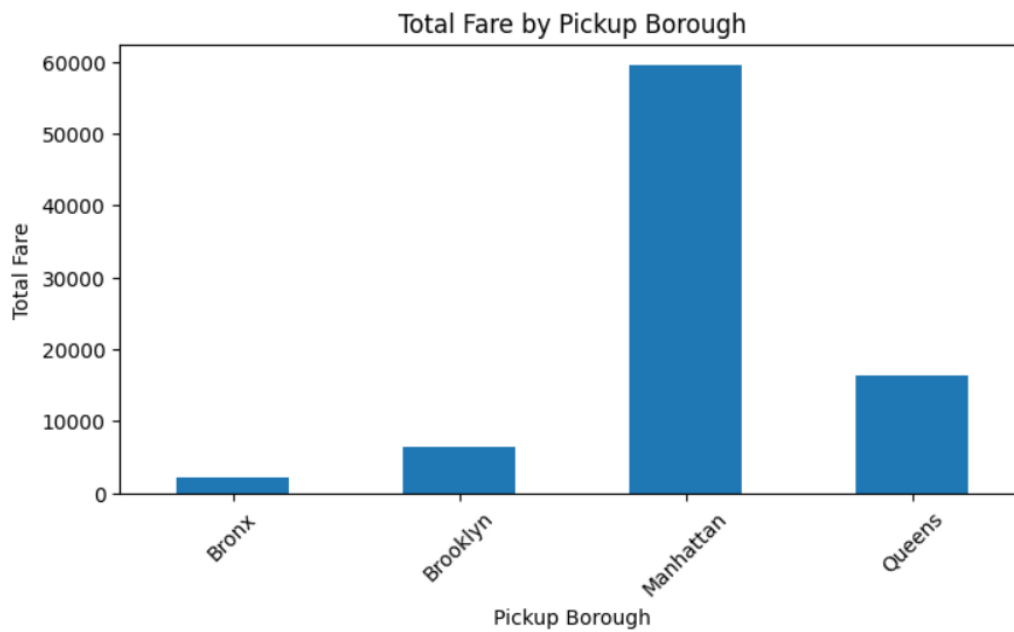
OUTPUT:



★ Bar Chart

Create a bar chart to show the total fare for each pickup_borough. Group the data by pickup_borough and sum the fare for each group.

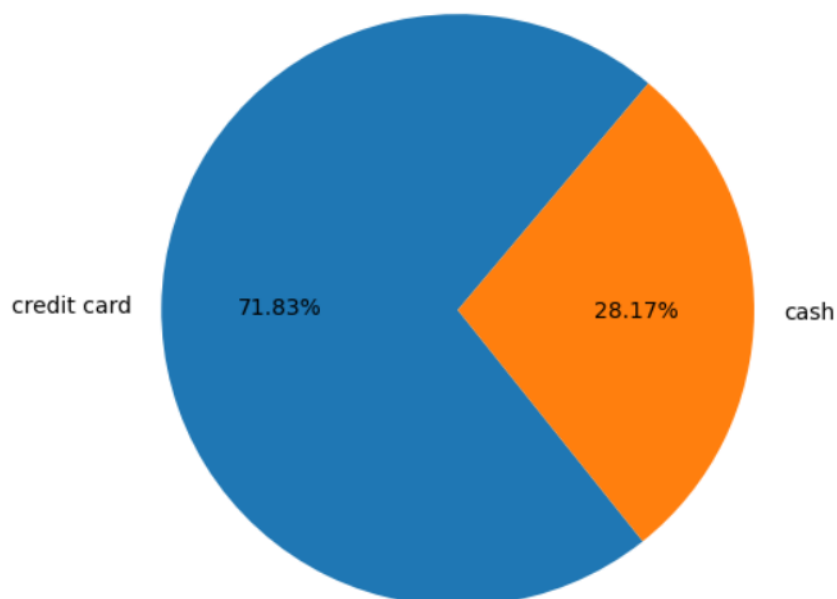
OUTPUT:



★ Pie Chart

Plot a pie chart showing the distribution of trips based on the payment method (credit card, cash, etc.). Each slice should represent the count of trips for a specific payment method.

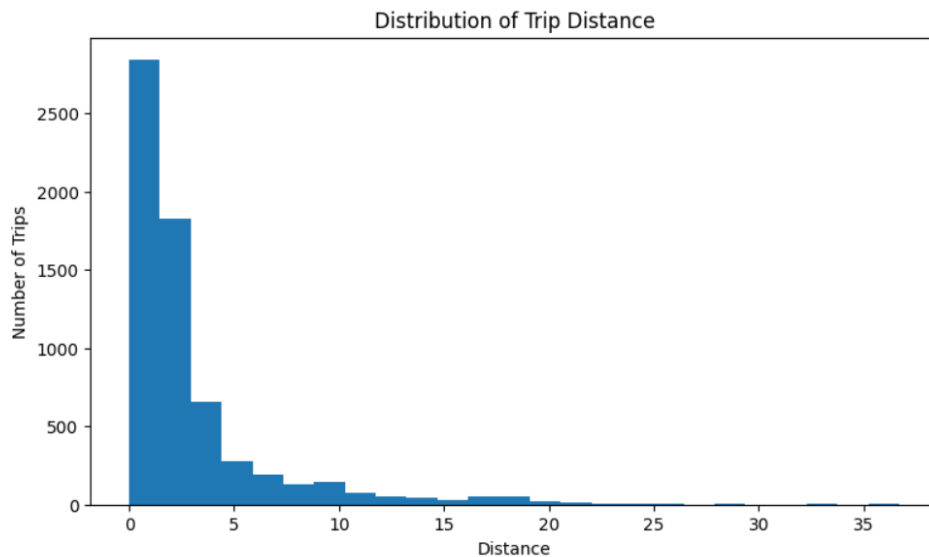
OUTPUT:



★ Histogram

Create a histogram to visualize the distribution of distance. Customize the number of bins for better granularity and ensure the plot is easy to interpret.

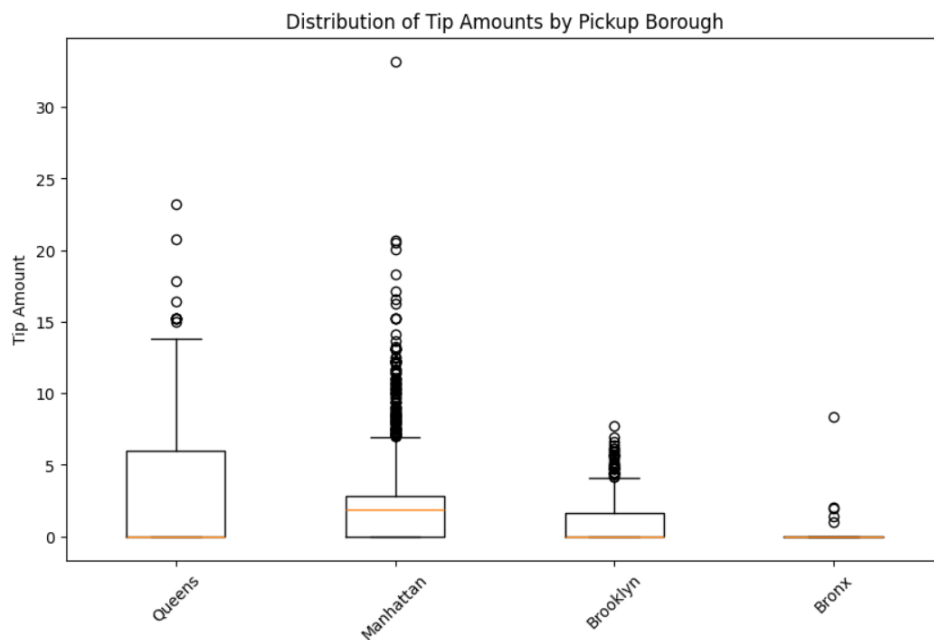
OUTPUT:



★ Box Plot

Plot a box plot to visualize the distribution of tip amounts for each pickup_borough. Use pickup_borough as the categorical axis and tip as the numeric axis.

OUTPUT:

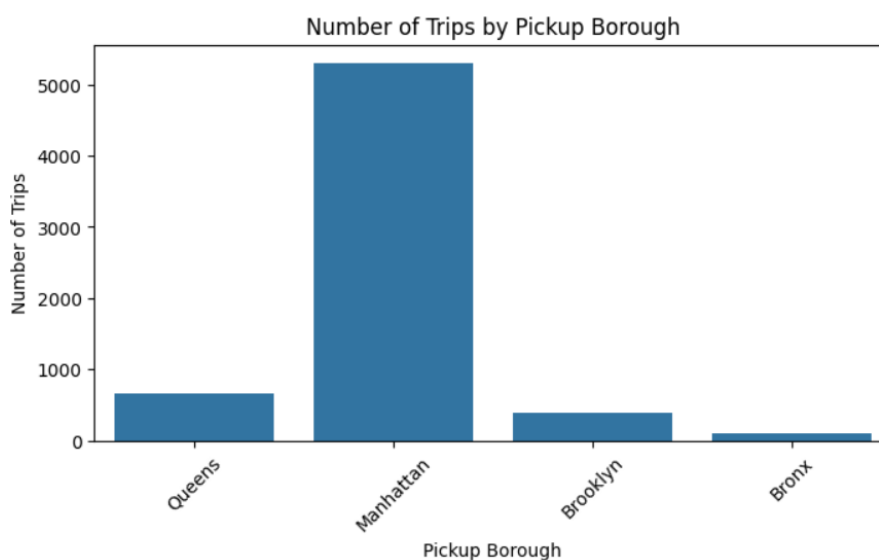


2) Visualizations using Seaborn:

★ Count Plot

Create a count plot to visualize the number of trips in each pickup_borough. The x-axis should represent the boroughs, and the y-axis should show the count of trips.

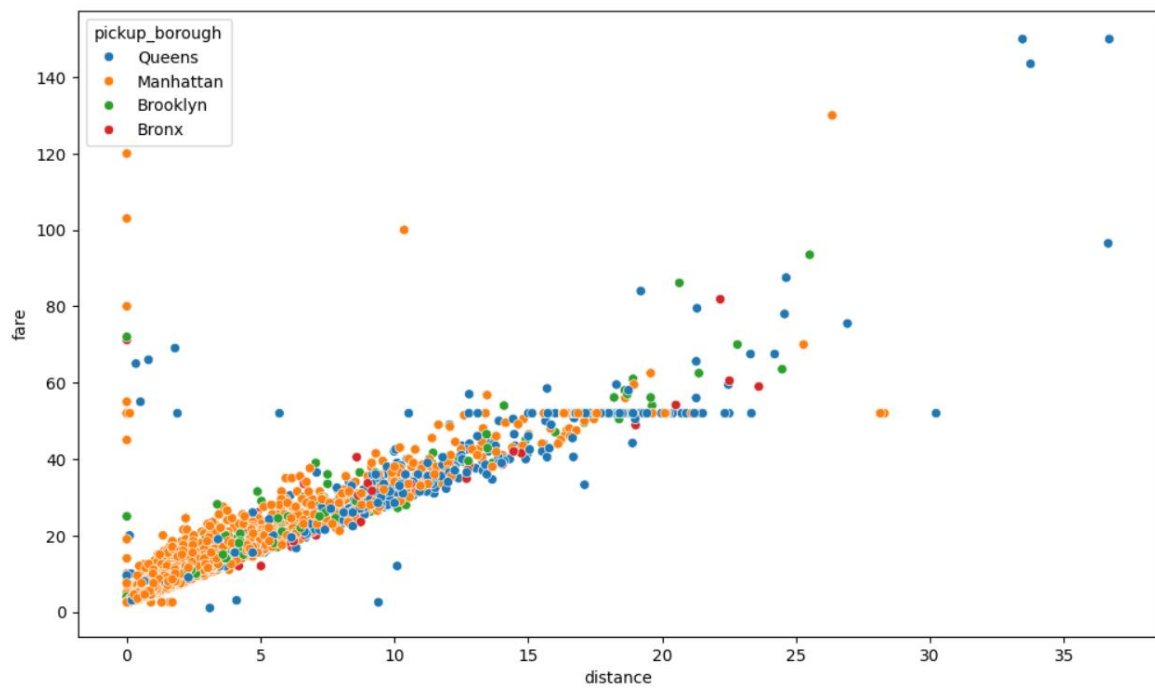
OUTPUT:



★ Scatter Plot

Plot a scatter plot to show the relationship between distance and fare. Use distance on the x-axis and fare on the y-axis to visualize any correlation. Color the points based on the pickup_borough to differentiate the trips by their respective boroughs.

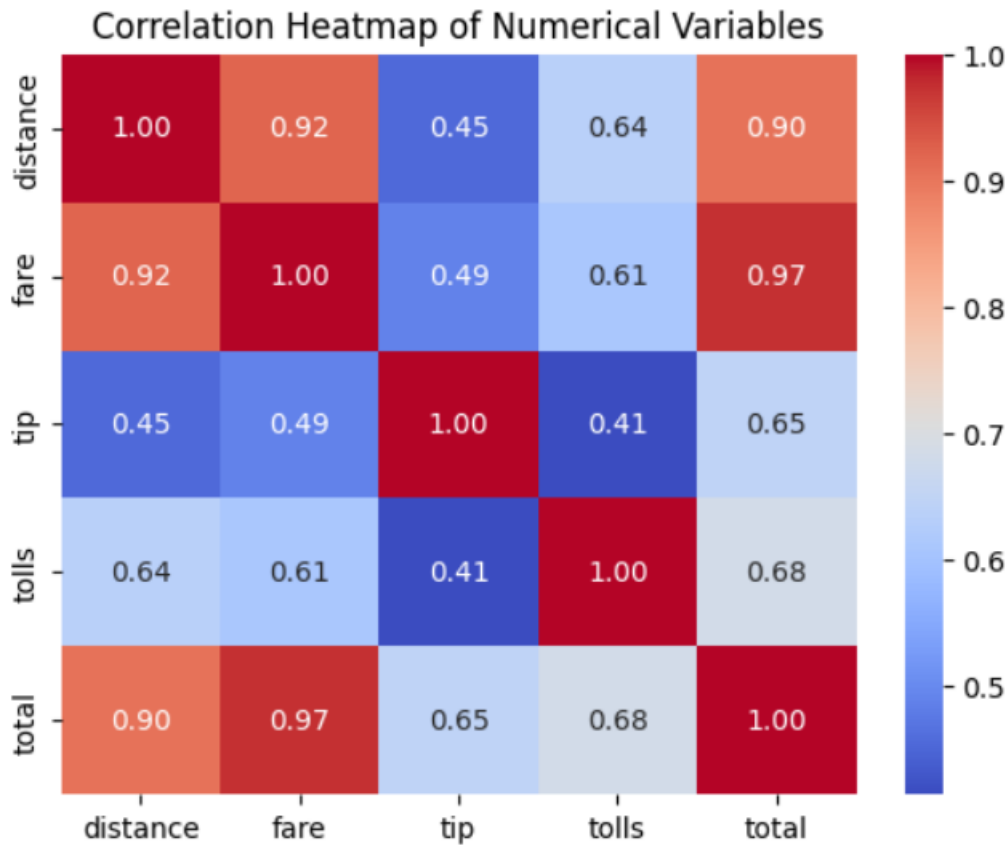
OUTPUT:



★ Heatmap

Plot a heatmap to visualize the correlation between numerical variables such as distance, fare, tip, tolls, and total. Use a correlation matrix to highlight the relationships.

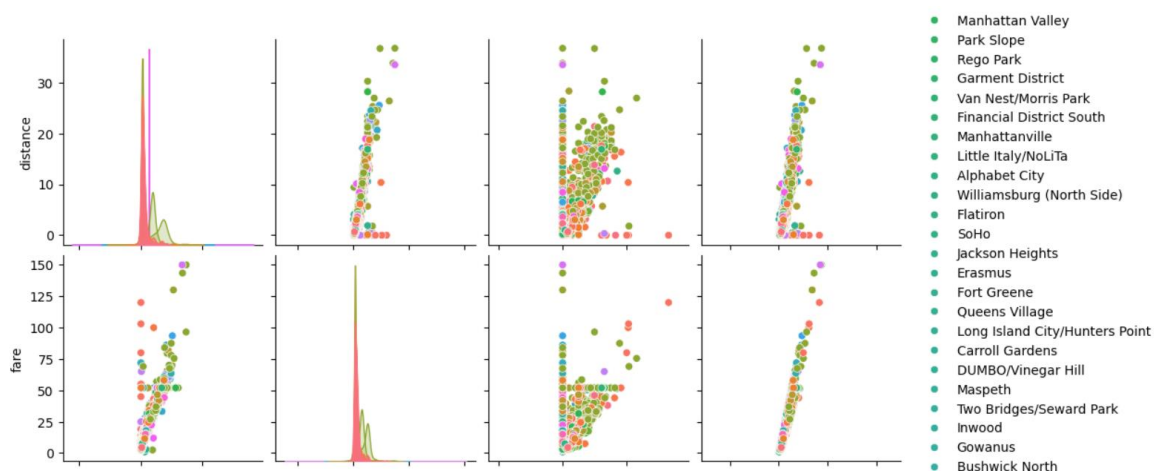
OUTPUT:

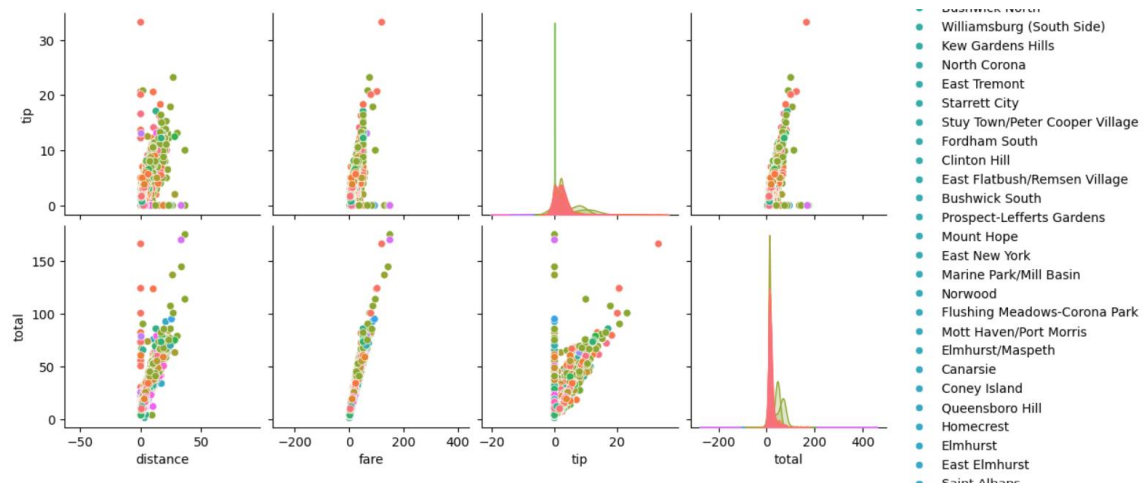


★ Pair Plot

Create a pair plot to visualize the pairwise relationships between distance, fare, tip, and total. Color the data points according to the pickup_zone column. to compare how different zones affect these variables.

OUTPUT:





★ Violin Plot

Plot a violin plot to show the distribution of fare for each payment method. Use the payment method as the categorical axis and fare as the numeric axis to visualize its distribution.

OUTPUT:

